

ded system. As compared to writing application software, writing embedded software is a different ball-game. You need to manipulate at bit level and use operands, operations, comments and mnemonics to carry out specific functionalities such as remote control. In this process, you may get acquainted with a number of productivity tools (makefiles), debuggers (gdb and kgdb) and profilers (gprof) of industrial importance.

What's next?

While dealing with the instruction set, try to customise yourself for the industry. Sharpen your skill sets by learning to use Gedit as a programmers editor, work with 'make' files and configure Gtkterm for serial data handling (UART) and understanding the AVR instruction set, embedded 'C' programming, variable scopes, qualifiers (volatile) and modifiers, program structures, pre-processor directives and determination of endianness (byte/bit ordering) of processors. Implementation knowledge of basic data structures like stacks and queues using above concepts may provide you an extra edge.

The instruction set is ready — what's next? Let's explore controller-specific features under the programming environment. At this point you need to know the aspects related to configuring ports and controlling their status. You have to assimilate memory and device interfacing, reading and writing EPROM as well as use of supported communication protocols and network protocols like SPI and I²C to get control over the system.

Try to dig a little deeper. Approach the critical parameters like interrupts, timers and boot loader in a realistic way, which I had already explained last month.

Institute watch

- **Think Labs** at Mumbai
Website: www.thinklabs.in
- **EFY Tech Centers** at Delhi, Bangalore
E-mail: efytech@efyindia.com

From an application perspective, knowledge of embedded 'C,' processor cores, communication protocols like SPI and I²C, network protocols, interfacing with analogue and digital signals, and RTOS is invaluable for those who will design and develop embedded systems.



—Sanjay Mittal, director, Yogasa Systems

The last frontier

Now, it's the time to address the real-world need. You may start RTOS porting and application development by using the skills you have gained so far. Try to proceed in a step-by-step manner. Review your real-time system concepts with respect to their need and constituting components. You may try to run a demo application to clarify the idea of RTOS porting and for better understanding of the firmware activities.

If you are confident up to that level, proceed to acquire some nice-to-have skillsets. Developing libraries and applications using inter-process communication mechanisms like mail boxes, priority inheritance under mutexes and semaphores can prove beneficial in the long run.

From classroom to industry

It is possible that your awareness of most of the above-mentioned terms is from a notional perspective only. But the industry needs something extra — the actual hands-on experience is indispensable. In fact, being in embedded systems means experiencing the great technological revolution. Your bookish knowledge might not be sufficient to catch the speed.

So be a go-getter, dig all the possibilities to get a practical exposure. At the same time, be up-to-date about the latest technological trends. Newspapers, technical magazines, various technical seminars and, last but not the

least, trade fairs may provide you with the opportunity to get an edge over your possible competitors.

You can learn about 'chip to ship' of an embedded system only after completing a proper project. Instead of depending only on your institutional projects, explore the possibilities to be a part of a live industrial project.

Various private- and public-sector organisations accommodate final-year and pre-final-year students in their live projects. I had discussed the relevant primary to-do steps in my article last month. Additionally, you may try to understand the project requirements in a better way with respect to design and documentation, development of prototype modules, coding and unit testing.

You may find various training institutes mushrooming all around. Before joining any course, be sure to compare the course curriculum with the skill sets discussed so far. Besides, you can always rely on vocational courses offered by the institutes of repute (see *Institute Watch*).

There has been a lot of speculation about the impact of the global recession on hiring in any industrial segment. As a budding professional, you could use this opportunity to improve your qualifications and inculcate additional skills that will come in handy once the market picks up again. ●

The author is a research analyst cum journalist at EFY